

C190

Introduction to Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A marathon runner who lives in Colorado breaks down glucose in her muscle cells for energy during a long run. Her coach tracks her split times with a GPS watch. Which organelle makes most of her ATP?

- A. Chloroplasts.
- B. Mitochondria.
- C. Ribosomes.
- D. Nucleus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A greenhouse in Arizona grows tomatoes under grow lights 16 hours a day. The owner sells at three farmers' markets. What raw materials do the plants use to make glucose in photosynthesis?

- A. Minerals from the soil.
- B. Glucose and oxygen.
- C. Oxygen and water.
- D. Carbon dioxide and water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A skin cell on the forearm of a 30-year-old lifeguard divides to heal a scrape. The lifeguard works 40 hours a week. How many chromosomes does each new cell have?

- A. 92
- B. 30
- C. 23
- D. 46

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A cell division ends with four genetically different cells, each with half the parent's chromosome number. Which process occurred?

- A.** Meiosis.
- B.** Binary fission.
- C.** Mitosis.
- D.** Mitosis, because all cell division halves chromosomes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Two pea plants, both heterozygous for flower color (Pp), are crossed in a school garden that gets full sun. Purple (P) is dominant over white (p). The class plants 200 seeds. What fraction of the offspring is expected to have white flowers?

- A.** 1/4
- B.** 1/2
- C.** None, because purple is dominant.
- D.** 3/4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A cross produces offspring in about a 1:1 ratio of dominant to recessive phenotype. What were the parents' genotypes most likely?

- A.** Aa × Aa
- B.** AA × aa
- C.** aa × aa
- D.** Aa × aa

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A lab at a coastal research institute studies a gene. The mRNA copy is made in the nucleus and then moves into the cytoplasm. The lab's microscope cost \$250,000. Where is the protein assembled from the mRNA?

- A. In the Golgi apparatus.
- B. In the mitochondria.
- C. At a ribosome.
- D. In the nucleus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hiker walking in hot weather starts to sweat, and her body temperature drops back toward normal. She is carrying a 20-pound pack. What type of control does this show?

- A. Muscle fatigue from the heavy pack.
- B. Negative feedback.
- C. Evolutionary adaptation.
- D. Positive feedback.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which is an example of positive feedback?

- A. Oxytocin release strengthening contractions during childbirth, which release more oxytocin.
- B. Insulin lowering high blood sugar.
- C. Any response that makes a person feel better.
- D. Shivering when body temperature drops.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

On an island that tourists visit each summer, a beetle population varies in shell color. Birds eat more light beetles than dark ones on the dark sand. The island is 12 miles long. What happens over many generations?

- A. The population shrinks because the island is small.
- B. All light beetles vanish in a single generation.
- C. Dark beetles become more common in the population.
- D. Individual light beetles darken their shells to survive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a lake, algae capture 20,000 kcal of energy. Small fish eat the algae, and bass eat the small fish. The lake is 40 feet deep. Using the 10% rule, about how much energy reaches the bass?

- A.** 16,200 kcal
- B.** 200 kcal
- C.** 2,000 kcal
- D.** 40 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

Tertiary consumers in a food chain receive 30 kcal. Using the 10% rule, about how much energy did the producers capture?

- A.** 3,000 kcal
- B.** 300 kcal
- C.** 30,000 kcal
- D.** 3 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A marathon runner ~~who lives in Colorado~~ breaks down glucose in her muscle cells for energy during a long run. ~~Her coach tracks her split times with a GPS watch.~~ Which organelle makes most of her ATP?

Target word: organelle for ATP **Decides:** Glucose breakdown with oxygen in an animal cell.

A	Inverted operation	Chloroplasts build glucose in plants; animal cells lack them.
B	Correct	Aerobic respiration makes most ATP in the mitochondria.
C	Confusion set	Ribosomes make proteins.
D	Adjacent category	The nucleus stores DNA and controls the cell but does not make most ATP.

2. Answer D Concept 2

A greenhouse ~~in Arizona~~ grows tomatoes under grow lights 16 hours a day. ~~The owner sells at three farmers' markets.~~ What raw materials do the plants use to make glucose in photosynthesis?

Target word: reactants of photosynthesis **Decides:** Light powers the conversion of CO₂ and water.

A	Adjacent category	Minerals support growth, but the carbon in glucose comes from the air.
B	Inverted operation	Those are the products of photosynthesis and the inputs of respiration.
C	Confusion set	Oxygen is released, not used, to make glucose.
D	Correct	Light energy turns CO ₂ and H ₂ O into glucose and O ₂ .

3. Answer D Concept 3

A skin cell on the forearm ~~of a 30-year-old lifeguard~~ divides to heal a scrape. ~~The lifeguard works 40 hours a week.~~ How many chromosomes does each new cell have?

Target word: chromosome count after body-cell division **Decides:** Skin repair uses mitosis.

A	Inverted operation	Doubles instead of keeping the count; DNA copies are split between the two cells.
B	Wrong stem number	The lifeguard's age.
C	Confusion set	That is the haploid number made by meiosis.
D	Correct	Mitosis keeps the full diploid number.

4. Answer A Concept 3 · transfer

A cell division ends with four genetically different cells, each with half the parent's chromosome number. Which process occurred?

Target word: reverse: result given, name the process **Decides:** Four haploid, varied cells.

A	Correct	Meiosis produces four genetically different haploid cells.
B	Adjacent category	Bacterial division that makes two identical cells.
C	Confusion set	Mitosis makes two identical cells with the full number.
D	Absolute language	Only meiosis halves the chromosome number.

5. Answer A Concept 4

Two pea plants, both heterozygous for flower color (Pp), are crossed ~~in a school garden that gets full sun.~~ Purple (P) is dominant over white (p). ~~The class plants 200 seeds.~~ What fraction of the offspring is expected to have white flowers?

Target word: recessive phenotype fraction **Decides:** Pp × Pp gives 1 PP : 2 Pp : 1 pp.

A	Correct	Only pp plants are white: 1 of 4 boxes in the Punnett square.
B	Confusion set	That is the heterozygous (Pp) fraction, which is purple.
C	Absolute language	Two carriers can each pass p, giving white offspring.
D	Inverted operation	That is the purple fraction.

6. Answer D Concept 4 • transfer

A cross produces offspring in about a 1:1 ratio of dominant to recessive phenotype. What were the parents' genotypes most likely?

Target word: solve for parents from the ratio **Decides:** Half the offspring show the recessive trait.

A	Confusion set	Gives a 3:1 ratio.
B	Adjacent category	All offspring are Aa and show the dominant trait.
C	Dropped element	Drops the dominant allele; all offspring would be recessive.
D	Correct	Half get A (dominant look), half get aa (recessive look).

7. Answer C Concept 5

A lab ~~at a coastal research institute~~ studies a gene. The mRNA copy is made in the nucleus and then moves into the cytoplasm. ~~The lab's microscope cost \$250,000.~~ Where is the protein assembled from the mRNA?

Target word: site of translation **Decides:** mRNA leaves the nucleus to be translated.

A	Confusion set	The Golgi modifies and ships proteins after they are made.
B	Adjacent category	Mitochondria make ATP.
C	Correct	Ribosomes read mRNA and link amino acids into protein.
D	Inverted operation	The nucleus is where transcription happens, the step before.

8. Answer B Concept 6

A hiker walking in hot weather starts to sweat, and her body temperature drops back toward normal. ~~She is carrying a 20-pound pack.~~ What type of control does this show?

Target word: type of feedback **Decides:** The response reverses the rise in temperature.

A	Camouflage	Answers the pack detail; it does not explain temperature control.
B	Correct	Sweating counteracts the change and returns temperature to the set point.
C	Adjacent category	Adaptation happens across generations, not within one hike.
D	Inverted operation	Positive feedback would push temperature further up.

9. Answer A Concept 6 • transfer

Which is an example of positive feedback?

Target word: reverse: find the amplifying loop **Decides:** The response makes the change stronger.

A	Correct	Each step amplifies the next until birth.
B	Confusion set	Reverses the change: negative feedback.
C	Absolute language	Feeling better does not define the feedback type.
D	Inverted operation	Brings temperature back up, which is negative feedback.

10. Answer C Concept 7

On an island ~~that tourists visit each summer~~, a beetle population varies in shell color. Birds eat more light beetles than dark ones on the dark sand. ~~The island is 12 miles long.~~ What happens over many generations?

Target word: outcome of selection **Decides:** Dark beetles survive and reproduce more.

A	Camouflage	Island size does not decide the color shift.
B	Absolute language	Selection shifts frequencies gradually.
C	Correct	Heritable dark color is passed on more often.
D	Confusion set	Individuals do not change inherited traits; populations change.

11. Answer B Concept 8

In a lake, algae capture 20,000 kcal of energy. Small fish eat the algae, and bass eat the small fish. ~~The lake is 40 feet deep.~~ Using the 10% rule, about how much energy reaches the bass?

Target word: energy at the third level **Decides:** Two transfers of 10% each.

A	Inverted operation	Subtracts 10% at each step instead of keeping 10%.
B	Correct	$20,000 \times 0.10 = 2,000$ to small fish; $\times 0.10 = 200$ to bass.
C	Dropped element	Stops after one transfer.
D	Wrong stem number	The lake depth.

12. Answer C Concept 8 • transfer

Tertiary consumers in a food chain receive 30 kcal. Using the 10% rule, about how much energy did the producers capture?

Target word: solve backward for producers **Decides:** Three transfers separate producers from tertiary consumers.

A	Confusion set	Treats tertiary consumers as two levels above producers.
B	Dropped element	Goes back only one level.
C	Correct	$30 \times 10 \times 10 \times 10 = 30,000$.
D	Inverted operation	Divides by 10 instead of multiplying when working backward.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Cellular respiration	Cells break down glucose with oxygen into CO ₂ and water, making most of their ATP in the mitochondria.	Thinking only plants or only chloroplasts make usable energy.	An animal cell using glucose for energy.
2	Photosynthesis	Plants use light energy to turn CO ₂ and water into glucose and release oxygen, in the chloroplasts.	Believing plant mass comes mainly from soil.	“Raw materials” for making glucose.
3	Mitosis vs. meiosis	Mitosis makes two identical diploid cells (46 chromosomes in humans); meiosis makes four genetically different haploid cells (23).	Halving chromosomes in ordinary body-cell division.	Body cell repair vs. egg or sperm formation.
4	Mendelian inheritance	A cross of two heterozygotes (Aa × Aa) gives about 3:1 dominant to recessive phenotypes and 1:2:1 genotypes.	Reporting the heterozygote fraction as the recessive fraction.	Two carriers and a question about the recessive trait.
5	Central dogma	DNA is transcribed to mRNA in the nucleus; mRNA is translated into protein at the ribosome.	Placing translation in the nucleus.	mRNA leaves the nucleus.
6	Homeostasis and feedback	Negative feedback reverses a change to restore a set point; positive feedback amplifies a change toward an endpoint.	Calling any response that brings relief positive feedback.	Does the response push the value back or push it further?
7	Natural selection	Selection acts on existing heritable variation; populations change over generations as better-suited traits leave more offspring.	Individuals change their traits to survive.	A predator removes one color more often.
8	Energy flow and the 10% rule	Only about 10% of energy passes to each higher trophic level.	Stopping after one step or subtracting 10% instead of keeping 10%.	Energy at producers and a question about a higher consumer.

Course scope (public description): This course is a foundational introduction to the biological sciences. The overarching theories of life from biological research are explored as well as the fundamental concepts and principles of the study of living organisms and their interaction with the environment. Key concepts include how living organisms use and produce energy; how life grows, develops, and reproduces; how life responds to the environment to maintain internal stability; and how life evolves and adapts to the environment.